

This week, you'll learn **Infrastructure as Code (IaC) & Configuration Management**. Terraform helps automate infrastructure provisioning, while Ansible helps configure and manage servers consistently. Together, they are essential for building repeatable DevOps environments.

Topic:- Infrastructure as Code & Configuration Management

- Infrastructure as Code (IaC)
- Terraform Fundamentals
- Terraform Providers
- Variables & Outputs
- Terraform State
- Terraform Modules
- terraform init, terraform plan, terraform apply, terraform destroy
- Ansible Fundamentals
- Ansible Inventory
- Ansible Playbooks
- Ansible Modules
- Configuration Management
- Terraform + Ansible Workflow

*Learning Approach

Focus on understanding **how infrastructure can be created and configured automatically** instead of manually setting up every server. Learn the difference between Terraform provisioning infrastructure and Ansible configuring it.

Keep these points in mind:

- ✓ Understand what Terraform and Ansible are responsible for.
- ✓ Learn the Terraform workflow before using apply.
- ✓ Understand variables, outputs, state, inventory, and playbooks.
- ✓ Always review changes before applying them.
- ✓ Test AI-generated IaC code before using it.

*Complete the following practical exercises:

- Install **Terraform** and **Ansible**.
- Configure a Terraform provider.
- Create a basic Terraform configuration.
- Use variables and outputs.
- Practice:
 - terraform init
 - terraform plan
 - terraform apply
 - terraform destroy
- Understand Terraform state.
- Create a basic Ansible inventory.
- Create an Ansible Playbook.
- Use Ansible to:
 - Install a package
 - Create a user
 - Start a service
- Document the difference between **Terraform and Ansible**.
- Prepare a short PDF explaining:
 - Infrastructure as Code
 - Terraform
 - Terraform State
 - Variables & Outputs
 - Modules
 - Ansible
 - Inventory
 - Playbooks
 - Terraform vs Ansible

Task Submission

- ✓ GitHub Repository Link
- ✓ Terraform .tf Files
- ✓ Ansible Inventory
- ✓ Ansible Playbook
- ✓ PDF Report
- ✓ Screenshots of Terraform Execution
- ✓ Screenshot of Ansible Playbook Execution

Task Submission Deadline: Wednesday (EOD)

Scenario

Imagine you are a **DevOps Engineer** responsible for preparing a new application server environment. The company wants the environment to be created consistently and configured automatically instead of performing the setup manually.

Use the infrastructure created during the **Weekly Task** and complete the following:

- Use Terraform to provision the required infrastructure.
- Use Ansible to configure the provisioned server.
- Configure the server with:
 - Required packages
 - Web server
 - Application user
 - Required service configuration
- Run the Ansible Playbook against the provisioned server.
- Verify that the server is configured correctly.
- Make one configuration change through Ansible and apply it again.
- Verify that the change is applied without manually configuring the server.
- Draw a simple workflow:

Terraform → Infrastructure → Ansible → Server Configuration → Application

Activity Submission

- ✓ Terraform Configuration Screenshot
- ✓ Ansible Execution Screenshot
- ✓ Configured Server Screenshot
- ✓ Workflow Diagram
- ✓ Short explanation of how Terraform and Ansible worked together

Activity Submission Deadline: Friday (EOD)

Use an AI Assistant such as **ChatGPT, Amazon Q, GitHub Copilot, Gemini, Claude, or Microsoft Copilot** as an **Infrastructure Automation Assistant**.

Explore how AI can be integrated into your IaC workflow:

- Ask AI to generate a basic Terraform configuration based on your requirements.
- Ask AI to review your Terraform code and identify possible improvements.
- Ask AI to generate an Ansible Playbook for a specific server configuration.
- Give AI a Terraform or Ansible error and ask it to identify possible causes.
- Ask AI to suggest improvements for making your IaC more reusable and maintainable.
- Compare the AI-generated solution with your own implementation.
- **Test and verify** the useful suggestions before applying them.

No separate AI submission is required.

The goal is to learn how AI can assist DevOps Engineers with infrastructure automation, configuration management, troubleshooting, and code improvement.

***DevOps Pro Tip**

Always run **terraform plan** before terraform apply. Review what Terraform wants to create, modify, or destroy before making changes to your infrastructure.

Understand how Terraform and Ansible work together rather than simply running commands or copying configuration files. Building strong IaC skills will help you create repeatable, scalable, and maintainable DevOps environments.

If you face any genuine issues, feel free to reach out before the submission deadline.